

Comparing Transformer Fine-Tuning, Multi-Task Learning, and LLM-as-Judge for Social Support Detection

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Abstract

Social support detection — identifying encouragement, solidarity, and care in social media text — requires fine-grained emotion and sentiment analysis beyond standard polarity classification. We participate in SSD-2026 at IberLEF 2026, a hierarchical shared task over 10,598 English and Spanish comments: Subtask 1 detects whether a comment is supportive; Subtask 2 identifies whether support targets an individual or a group; and Subtask 3 classifies the targeted community (Nation, LGBTQ, Black Community, Women, Religion, or Other). We compare five approaches: TF-IDF baselines, FastText deep learning models, domain-adapted transformer fine-tuning (BERTweet, RoBERTuito, XLM-R), multi-task learning (XLM-Twitter), and LLM-as-judge prompting with GPT-5.4-mini using emotion-aware few-shot examples. Our best submissions achieve official CodaBench scores of 0.8231 for English and 0.8785 for Spanish.

Keywords

social support detection, transformer fine-tuning, multi-task learning, LLM-as-judge, BERTweet, cross-lingual NLP, shared task

1. Introduction

Natural language processing (NLP) has driven rapid progress in understanding human language at scale, enabling automated analysis of text that would be infeasible by manual inspection alone [1, 2]. Social media platforms — Twitter, Reddit, YouTube, Instagram — have become primary channels for public discourse, generating billions of messages daily that reflect opinions, emotions, and interpersonal dynamics in real time [3, 4]. This volume of user-generated text presents both an opportunity and a challenge for NLP: while the data is abundant, it is noisy, informal, and context-dependent.

Within this landscape, the intersection of social support and social media has received growing attention from both the social sciences and the NLP community [5, 6]. Social support — acts of encouragement, solidarity, care, or help directed at others — plays a critical role in mental health, community resilience, and crisis response [7, 8]. Detecting such supportive language in social media enables applications ranging from monitoring community wellbeing to identifying vulnerable populations in need of intervention [9, 10]. However, this task is distinct from sentiment analysis: a comment can be emotionally neutral yet deeply supportive, emotionally charged yet not supportive of anyone, and the *target* of support (an individual versus a community) matters for downstream applications.

The gap. Despite this importance, most NLP research on social media text has focused on detecting *negative* phenomena — hate speech, toxicity, offensive language [11, 12] — while *prosocial* language remains comparatively understudied. Similarly, sentiment analysis often reduces affect to positive or negative polarity, whereas supportive sentiment requires detecting directed care, encouragement, and solidarity. Existing work on social support detection has been limited in scope: early studies relied on lexicon-based and feature-engineering methods applied to specific health forums [8], and more recent transformer approaches have been evaluated on small English-only datasets [13, 14]. Key open questions remain: How do domain-adapted transformer models compare to general-purpose multilingual models and large language models (LLMs) for this task? Does multilingual training help in data-scarce subtasks? Can LLMs complement fine-tuned classifiers, particularly for underrepresented classes?

Social Support Detection at IberLEF 2026 (SSD-2026)

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The shared task. The Social Support Detection shared task at IberLEF 2026 (SSD-2026) [15] addresses these gaps by formalizing social support detection as a three-level hierarchy: Subtask 1 determines whether a comment is supportive; Subtask 2 classifies support as targeting an individual or a group; and Subtask 3 identifies the specific community (Nation, Other, LGBTQ, Black Community, Religion, or Women). The task covers both English (7,998 training samples) and Spanish (2,600 samples) social media text, with pronounced class imbalance — as few as 15 training examples for the Religion class.

Our contribution.

- We systematically compare five modeling paradigms for social support detection across English and Spanish: TF-IDF baselines, FastText neural models, single-task transformers, multi-task learning, and LLM-as-judge prompting.
- We frame social support detection as emotion and sentiment analysis of prosocial language, using emotion-aware prompting to separate directed support from undirected emotional reactions.
- We show that multilingual training improves English minority-class performance, with XLM-R improving Subtask 3 by 12.5 F1 points over English-only training.
- We find that hybrid LLM+transformer verification is limited by anchoring bias: the LLM over-agrees with provided pre-labels instead of independently correcting errors.
- Our official CodaBench scores are 0.8231 for English (three-way tie) and 0.8785 for Spanish.

2. Related Work

Social support detection. Early NLP work on social support drew from social science frameworks distinguishing informational, emotional, and instrumental support [7], applying lexicon-based and feature-engineering methods to health forums and Reddit communities [8, 9]. Neural approaches later showed that deep models capture subtle supportive cues that surface-level features miss [10]. Most directly related to our work, Burnap et al. [14] formalize social support detection as the same three-task hierarchy used in SSD-2026, and González et al. [13] benchmark transformer approaches on this setup, with RoBERTa [16] reaching macro F1 of 0.78–0.84 across subtasks. Rosales et al. [17] extend the framework to Spanish social media, demonstrating cross-lingual applicability. Adjacent work on hope speech [18] and empathetic dialogue [19] further shows that positively directed prosocial language is tractable with transformer-based classifiers [20].

Transformer models for social media. BERT [2] and its variants have become the standard backbone for social media classification. For cross-lingual scenarios, Conneau et al. [21] introduced XLM-R, a RoBERTa model pretrained on 2.5TB of filtered Common Crawl data across 100 languages, which achieves strong zero-shot cross-lingual transfer. Domain-adapted models offer additional gains on informal text: Nguyen et al. [22] pretrained BERTweet on 850 million English tweets, and Pérez et al. [23] released RoBERTuito on over 500 million Spanish tweets, both outperforming general-purpose models on Twitter classification tasks [3, 4].

Target identification and class imbalance. The hierarchical structure of SSD-2026 — detect, then classify the target — mirrors shared tasks on offensive language such as OffenseEval [11, 24] and HatEval [12], which require identifying whether content targets an individual or a group across English and Spanish Twitter. Work on community-targeted classification [25] has shown that minority groups are consistently harder to classify, a finding directly applicable to Subtask 3 where Women (19 examples) and Religion (15 examples) are severely underrepresented. We address this through inverse-frequency class weighting [13], a lighter-weight alternative to oversampling [26] and focal loss [27].

3. Task and Dataset Description

3.1. Task Definition

SSD-2026 is organized as a hierarchical three-subtask shared task on English and Spanish social media comments. A comment that expresses encouragement, care, admiration, help, or solidarity toward someone is labeled as *Supportive*; all others, including comments that promote violence or harm even if superficially positive, are labeled *Not Supportive*.

Subtask 1 — Support Detection (Binary). Given any social media comment, predict whether it is Supportive or Not Supportive.

Subtask 2 — Target Type Identification. Given a comment already labeled Supportive, predict whether the support targets an *Individual* or a *Group*.

Subtask 3 — Targeted Group Classification. Given a Supportive comment that targets a group, predict the specific community from six classes: Nation, Other, LGBTQ, Black Community, Religion, and Women.

3.2. Dataset

The official training set consists of 7,998 English-language and 2,600 Spanish-language social media comments drawn from diverse platforms. Table 2 summarizes the label distributions for both languages across all three subtasks.

The dataset exhibits pronounced class imbalance throughout. For Subtask 1, only 22.4% of English comments are labeled Supportive (1,795 out of 7,998), while Spanish shows a similar ratio at 22.7% (590 out of 2,600), motivating the use of class-weighted loss during training. Subtask 2 is applied exclusively to Supportive comments: in English, the majority target a group (1,450) rather than an individual (345), yielding a roughly 4:1 imbalance; Spanish shows a comparable 3.4:1 ratio (456 vs. 134). Subtask 3 further partitions the group-supportive comments into six community categories. In English, Nation (786) and Other (416) dominate while Women (19) and Religion (15) are severely under-represented. The Spanish distribution is strikingly different: LGBTQ (196) is the largest class, while Nation (47) and Black Community (24) are the rarest, reflecting distinct social media discourse contexts across languages.

Table 1 gives representative training examples. These cases illustrate why the task goes beyond polarity: emotional reactions, political commentary, and directed support can use overlapping vocabulary while requiring different labels.

Table 1

Representative examples from the English training set. Text is shortened for readability.

Text	T1	T2	T3
“happy see smiling child god bless family helped”	Supportive	Individual	—
“free free palestine ... stand palestine”	Supportive	Group	Nation
“profound showing yet making sad want cry poor babies”	Non-Supportive	—	—
“black lives matter stupid movement ...”	Non-Supportive	—	—
“stay strong stay proud”	Supportive	Group	LGBTQ

Figure 1 visualizes the label distribution for Subtask 1 across both languages, highlighting the roughly 4:1 Non-Supportive to Supportive ratio in English and a more balanced split in Spanish. Figures 2 and 3 show the corresponding distributions for Subtasks 2 and 3.

3.3. Text Length Analysis

Figure 4 shows the word-count distribution by support label for both languages. In English, Supportive comments are slightly longer on average (mean = 19.6 words) than Non-Supportive ones (mean = 18.4).

Table 2

Label distribution across all subtasks for English and Spanish training data.

Subtask	Label	English	Spanish
1 – Support	Non-Supportive	6,203	2,010
	Supportive	1,795	590
2 – Target	Group	1,450	456
	Individual	345	134
3 – Community	Nation	786	47
	Other	416	81
	LGBTQ	123	196
	Black Community	91	24
	Women	19	48
	Religion	15	60

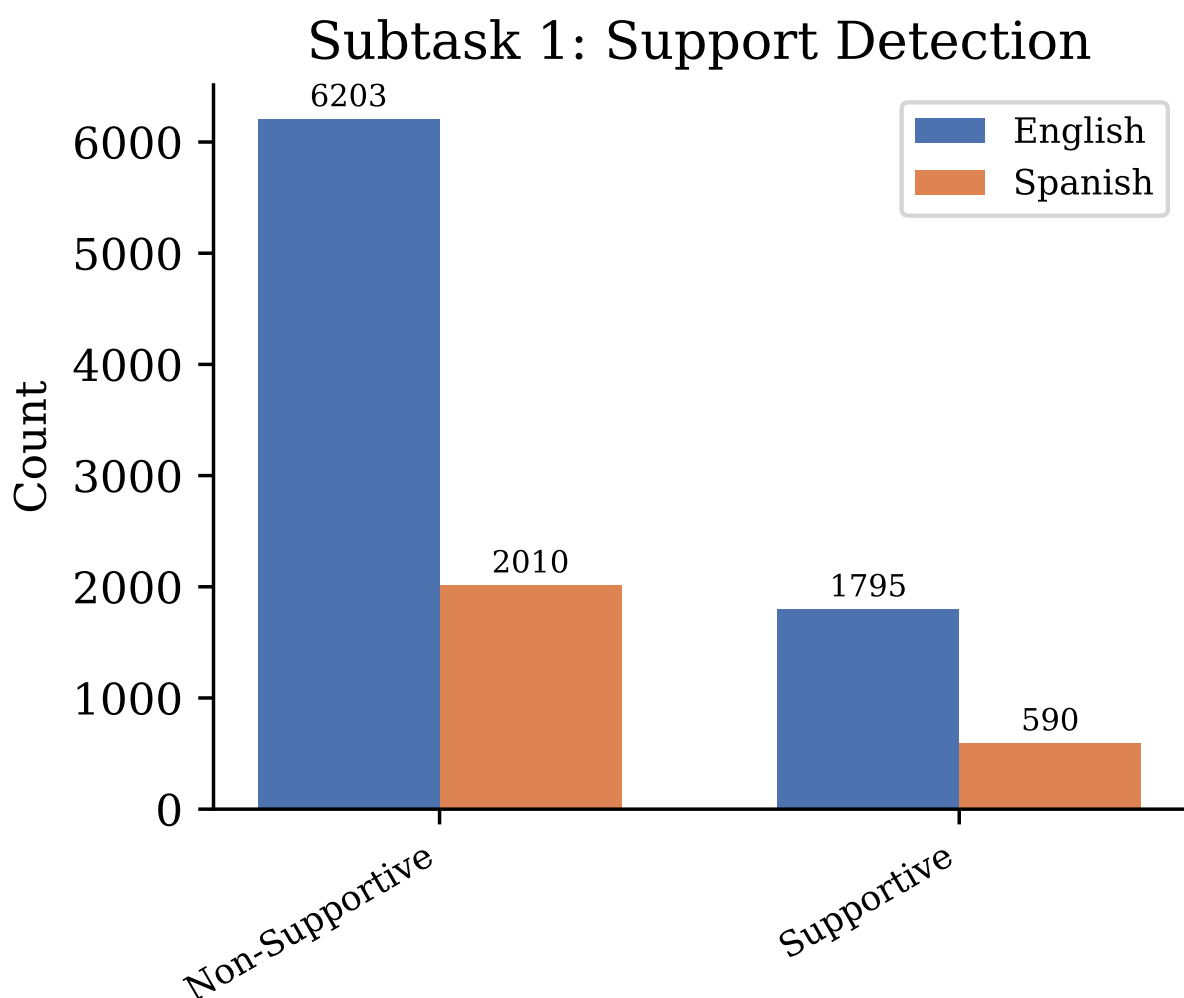


Figure 1: Subtask 1 label distribution across English and Spanish.

Spanish comments are generally longer but with higher variance (Supportive: mean = 28.7, std = 50.9; Non-Supportive: mean = 24.4, std = 39.2), reflecting a wider range of comment lengths in the Spanish social media sources. Both distributions are heavily right-skewed, with the majority of comments under 40 words.

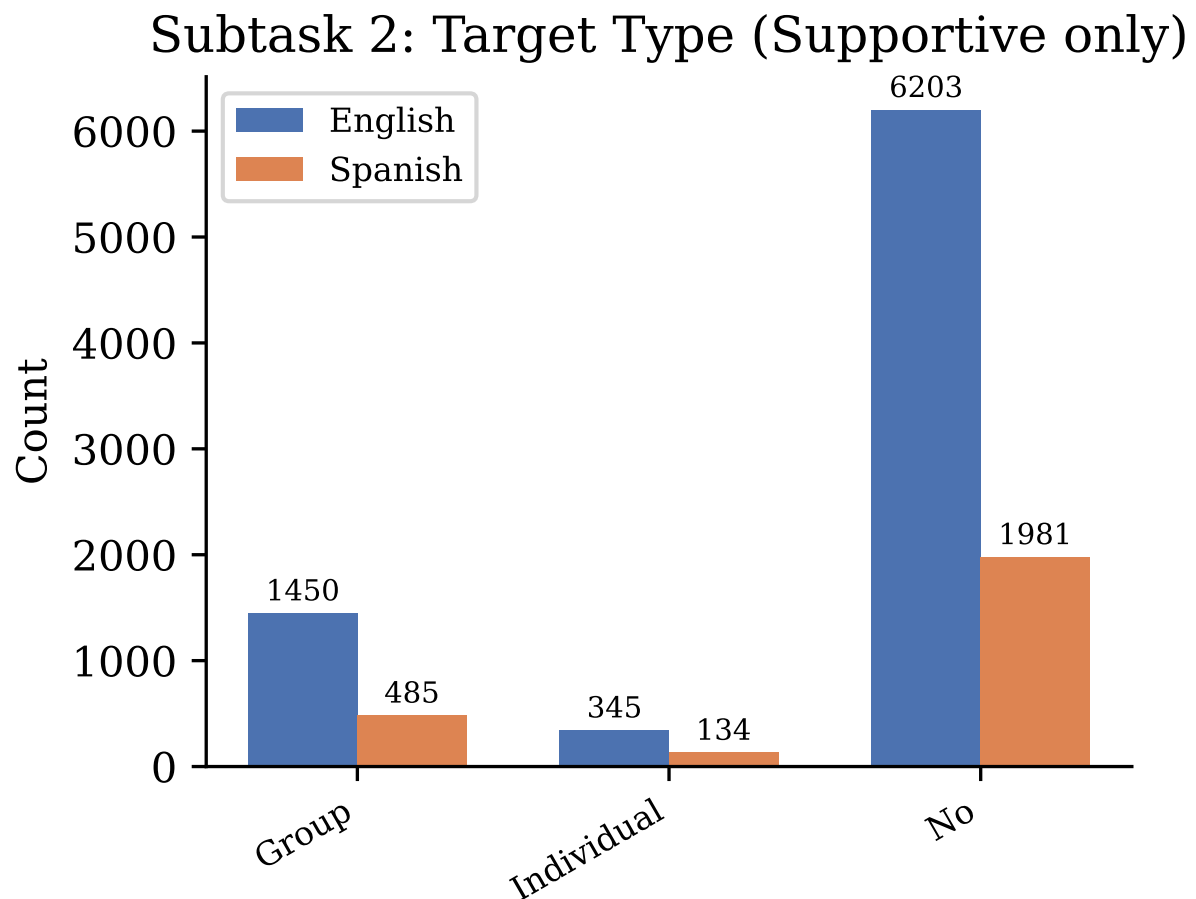


Figure 2: Subtask 2 label distribution (Supportive comments only).

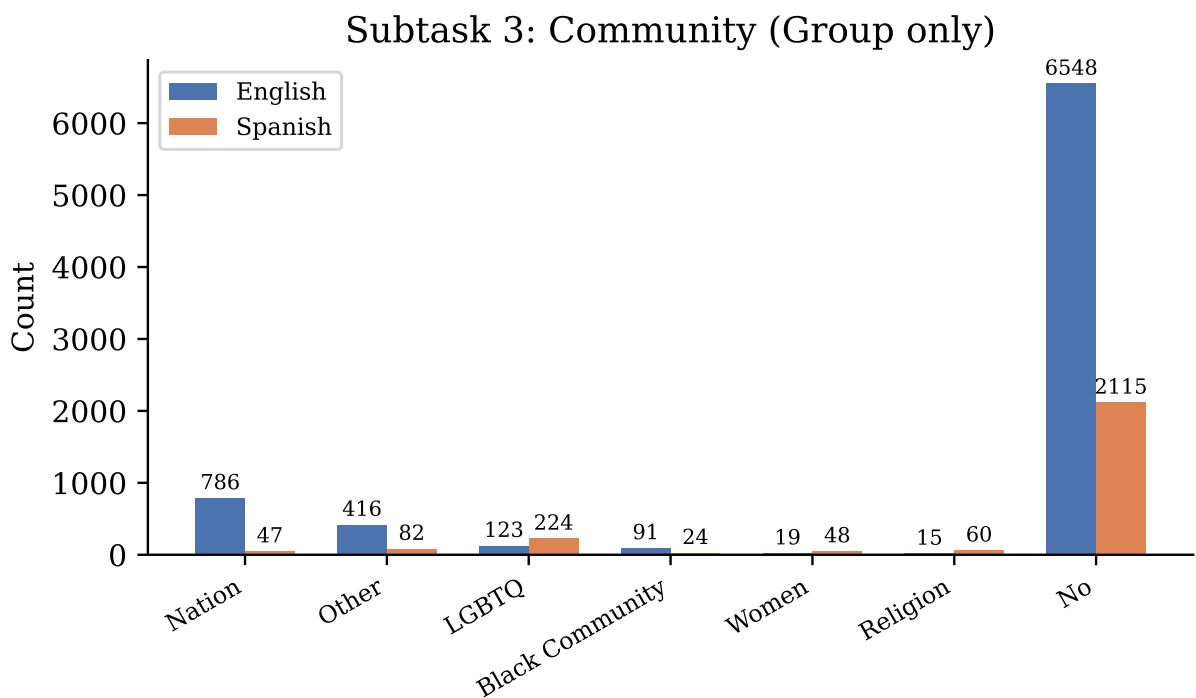


Figure 3: Subtask 3 label distribution (Group-supportive comments only).

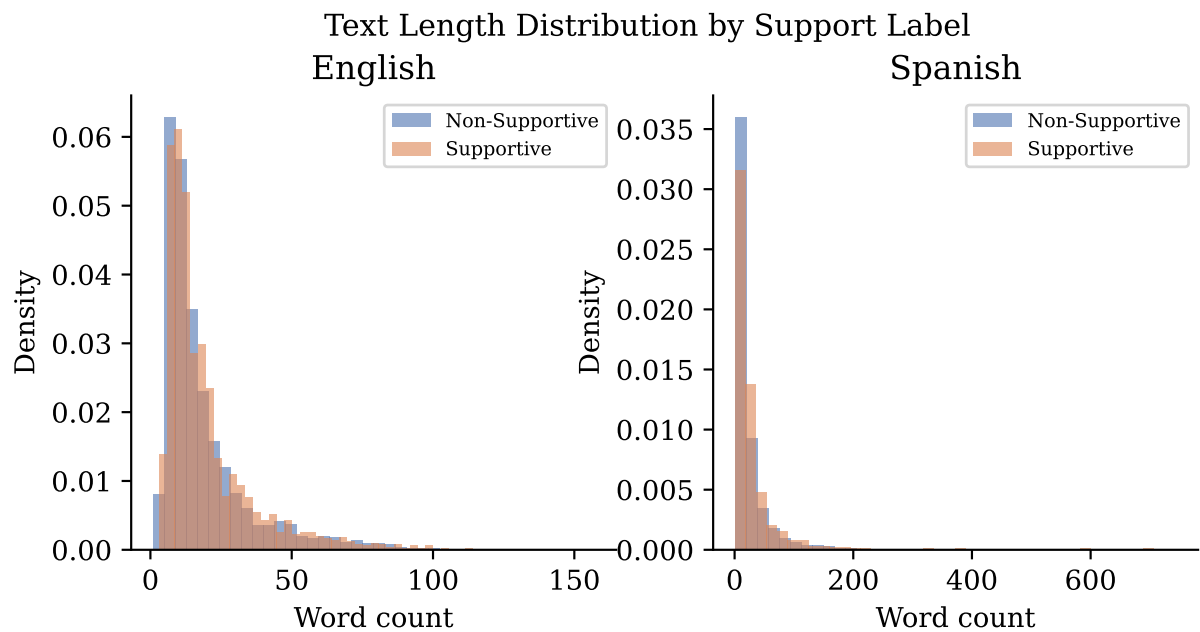


Figure 4: Text length (word count) distribution by support label.

3.4. Lexical Analysis

To understand the vocabulary distinguishing Supportive from Non-Supportive text, we extract the most frequent bigrams and trigrams per class after removing English stop words.

Figure 5 shows the top 15 bigrams for each Subtask 1 class in English. Supportive comments are characterized by bigrams expressing solidarity and appeals to faith (e.g., “free palestine,” “god bless,” “ya allah,” “black lives”), while Non-Supportive comments share some of the same topics (“lives matter,” “black people”) but in a more argumentative or commentary context (“gon na,” “white people,” “feel like”). Figure 6 shows the corresponding trigrams, where “black lives matter” dominates both classes but with different framing, and the Supportive class additionally features Palestine-related solidarity phrases (“free free palestine,” “brothers sisters palestine”).

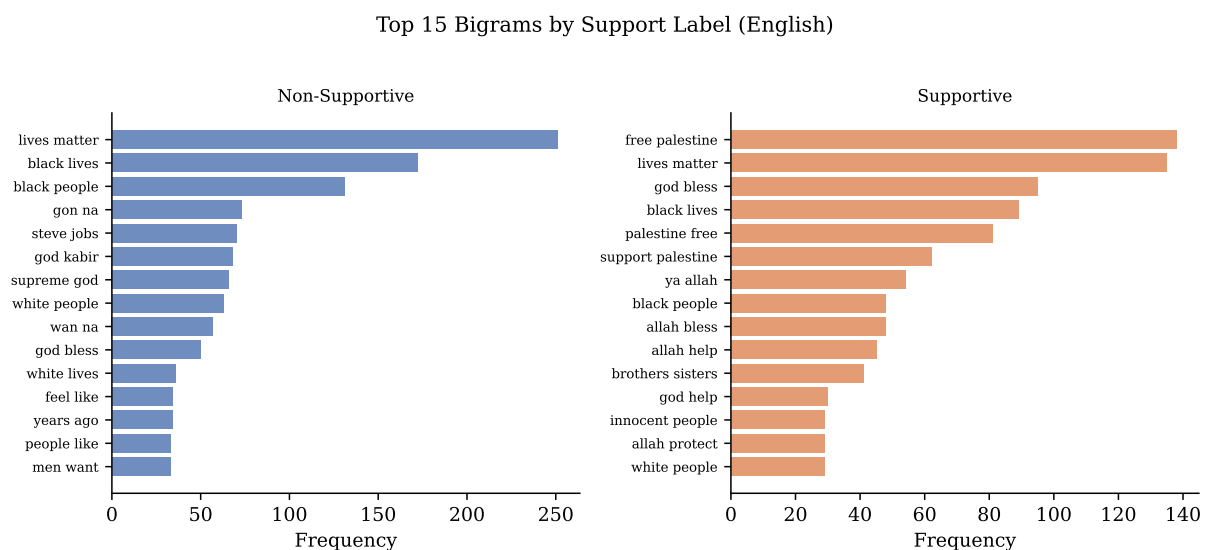


Figure 5: Top 15 bigrams by Subtask 1 label (English, stop words removed).

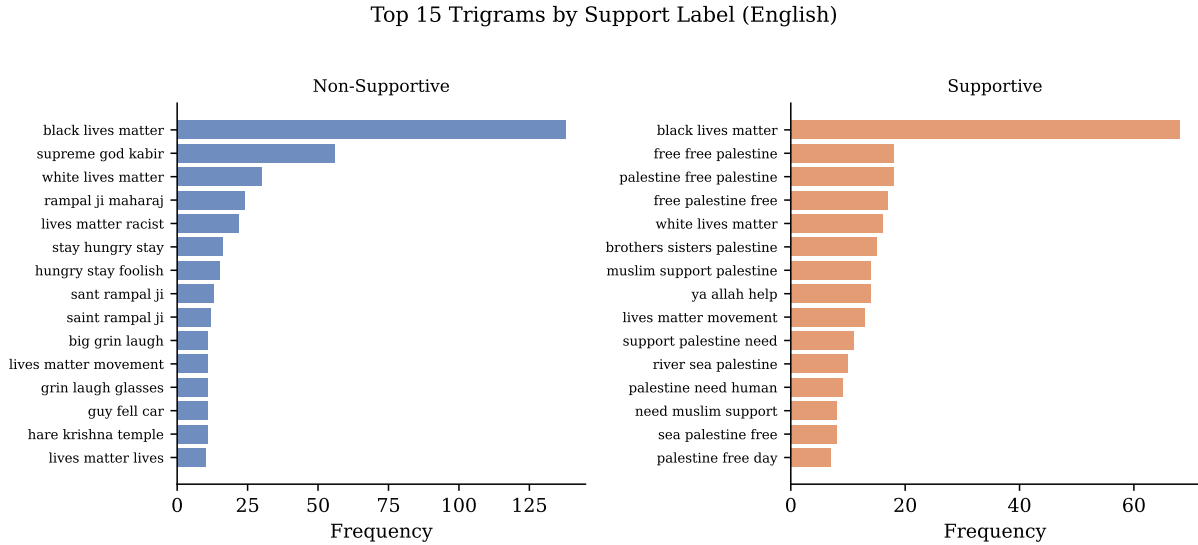


Figure 6: Top 15 trigrams by Subtask 1 label (English, stop words removed).

4. Methodology

4.1. Pipeline and Evaluation Protocol

We treat SSD-2026 as a hierarchical inference problem. At prediction time, Subtask 1 is applied to every comment; Subtask 2 is applied only to comments predicted as Supportive; and Subtask 3 is applied only to comments predicted as both Supportive and Group-targeted. This sequential design matches the shared task definition and prevents irrelevant Subtask 2 and Subtask 3 labels from being assigned to Non-Supportive comments.

All internal experiments are evaluated with macro-averaged F1, the official shared task metric, computed separately for each subtask. For model selection, we use the best validation macro F1 and report the best single seed together with the mean and standard deviation over three seeds where available. Official leaderboard results are reported separately in Section 6 because CodaBench scores aggregate the official test performance for a language submission and are not interchangeable with per-subtask internal validation F1.

4.2. Data Splits

The full training set is partitioned into train, validation, and test subsets at an 80/10/10 ratio using stratified sampling. For Subtask 2, only Supportive-labeled examples are retained, yielding approximately 1,436 / 179 / 180 English examples. For Subtask 3, only Group-targeted examples are retained, reducing to approximately 1,160 English training examples across six categories. The same stratified procedure is used for Spanish and multilingual experiments.

4.3. Training Configuration

Table 3 summarizes the key hyperparameters for each modeling approach. All fine-tuned transformer models use the AdamW optimizer with weight decay 0.01, linear warmup over the first 10% of training steps followed by linear decay, gradient clipping at max ℓ_2 norm of 1.0, and inverse-frequency class-weighted cross-entropy loss. Each transformer configuration is evaluated across 3 random seeds (13, 42, 77). Deep learning models use 300-dimensional pretrained FastText embeddings. Traditional ML models use combined word-level and character-level TF-IDF (unigrams + character 3–5-grams) with up to 50,000 features, augmented by handcrafted features (text length, punctuation counts, uppercase

ratio). Experiments were implemented in PyTorch and run on available CUDA or Apple MPS devices depending on the host machine.

Table 3

Hyperparameter summary for each modeling approach. LR = learning rate, BS = batch size, Pat. = early stopping patience.

Approach	Model	LR	BS	Max Len	Epochs	Pat.	Other
Traditional ML	LR / SVM	—	—	—	—	—	TF-IDF 50k features
Deep Learning	TextCNN	10^{-3}	64	128	20	5–7	Dropout 0.3, filters [2,3,4,5]
	BiLSTM	10^{-3}	64	128	20	5–7	Dropout 0.3, hidden dim 256
Single-Task FT	BERTweet	2×10^{-5}	32	128	5	2–3	Warmup 10%, wd = 0.01
	RoBERTuito	2×10^{-5}	32	128	5	2–3	Warmup 10%, wd = 0.01
	XLM-R	2×10^{-5}	32	128	5	2–3	Warmup 10%, wd = 0.01
Multi-Task	XLM-T (ph. 1)	10^{-4}	16	128	3	—	Heads only, grad accum 2
	XLM-T (ph. 2)	2×10^{-5}	16	128	var	3	End-to-end, $\lambda_{1:3} = 1$
LLM Judge	GPT-5.4-mini	—	—	—	—	—	Temp = 0.0, 26 few-shot ex.

5. System Description

5.1. Overall Approach

We explore five modeling paradigms for each subtask, summarized in Figure 7:

1. **Traditional ML:** Logistic Regression and SVM with TF-IDF features augmented by handcrafted features (text length, punctuation counts, emoji density).
2. **Deep Learning:** TextCNN and BiLSTM with pretrained FastText embeddings.
3. **Single-Task Transformers:** BERTweet (English), RoBERTuito (Spanish), and XLM-R (both languages plus multilingual training).
4. **Multi-Task Learning:** A shared XLM-Twitter encoder with three task-specific classification heads, trained jointly on all subtasks.
5. **LLM-as-Judge:** GPT-5.4-mini with a carefully engineered combined few-shot prompt classifying all three subtasks simultaneously.

For the competition submission, we adopt a *pipeline* strategy: three independent models are applied sequentially, each consuming only the subset of examples relevant to its subtask. The best model for each subtask is selected based on validation macro F1.

Approach 1: TF-IDF + LR / SVM
Approach 2: FastText + TextCNN / BiLSTM
Approach 3: BERTweet / RoBERTuito / XLM-R (single-task)
Approach 4: XLM-Twitter (multi-task, 3 heads)
Approach 5: GPT-5.4-mini (few-shot LLM-as-judge)

Figure 7: Overview of the five modeling paradigms explored.

5.2. Text Preprocessing

We apply different preprocessing pipelines depending on the model family:

- **Traditional ML:** Aggressive preprocessing — lowercasing, URL and mention removal, emoji-to-text conversion, slang expansion, and lemmatization with spaCy.

- **Deep Learning:** Moderate preprocessing — lowercasing, URL and mention removal, emoji-to-text conversion.
- **Transformers and LLM:** Minimal preprocessing — Unicode normalization (NFKC), emoji-to-text conversion via the `emoji` library, and whitespace collapsing. We let the subword tokenizer handle the rest, preserving casing and punctuation that carry semantic signal in social media.

5.3. Traditional ML Baselines

We train Logistic Regression (LR) and linear SVM classifiers on combined TF-IDF features (word unigrams + character 3–5-grams, max 50,000 features) augmented with handcrafted features including text length, exclamation mark count, question mark count, and uppercase ratio.

5.4. Deep Learning Models

We experiment with TextCNN [3] using filter sizes [2, 3, 4, 5] and BiLSTM with 256-dimensional hidden states, both initialized with 300-dimensional FastText embeddings. Both models use dropout (0.3), class-weighted cross-entropy loss, and early stopping on validation macro F1 with patience of 5–7 epochs.

5.5. Single-Task Transformer Fine-Tuning

Each subtask model fine-tunes a pretrained transformer encoder with a linear classification head over the [CLS] token. We evaluate three encoders:

- **BERTweet** [22]: Pretrained on 850M English tweets; used for English subtasks.
- **RoBERTuito** [23]: Pretrained on 500M+ Spanish tweets; used for Spanish subtasks.
- **XLM-R** [21]: Multilingual RoBERTa pretrained on 100 languages; evaluated on English-only, Spanish-only, and combined multilingual training data.

The shared training settings, including maximum sequence length, class weighting, early stopping, and multi-seed evaluation, are detailed in Section 4.

5.6. Multi-Task Learning

We train a multi-task model with a shared XLM-Twitter (`cardiffnlp/twitter-xlm-roberta-base`) encoder and three task-specific linear classification heads. Training proceeds in two phases: (1) a warm-up phase (3 epochs) training only the classification heads with learning rate 10^{-4} , followed by (2) end-to-end fine-tuning of all parameters with learning rate 2×10^{-5} . The total loss is a weighted sum of the three task-specific cross-entropy losses with equal weights ($\lambda_1 = \lambda_2 = \lambda_3 = 1$). Gradient accumulation (2 steps) with batch size 16 yields an effective batch size of 32.

5.7. LLM-as-Judge

We use GPT-5.4-mini as a zero-resource classifier via the OpenAI API. A single combined prompt instructs the model to classify all three subtasks at once, returning a JSON object with keys `task1`, `task2`, and `task3`. The prompt includes:

- Detailed task definitions with explicit boundary conditions (e.g., “admiration directed at a person is Supportive; general praise of qualities is Non-Supportive”).
- 26 few-shot examples drawn from the training data, including deliberately tricky cases (emotional reactions to tragedy, sports fan debate, religious preaching, political opposition).
- Temperature set to 0.0 for deterministic output.

5.8. LLM Prompt Design

The final prompt combines a system role, task definitions, emotion-aware boundary rules, 26 few-shot examples, and a strict JSON output schema. The key design decision was to distinguish undirected emotional reactions from directed support: comments such as “tears come to my eyes” express emotion but are Non-Supportive unless they include encouragement, care, or solidarity toward a person or group.

System: You are a social media analyst specializing in emotion and sentiment analysis.
Classify the comment for three tasks simultaneously and return JSON.

Task 1: Is the comment Supportive or Non-Supportive?

Supportive: encouragement, solidarity, care, help, or admiration directed at a person/group.

Non-Supportive: analytical, argumentative, harmful, or emotional reaction without directed support.

Task 2: If Supportive, is the target an Individual or Group?

Task 3: If Group, choose Nation, LGBTQ, Black Community, Women, Religion, or Other.

Return: {"task1": "...", "task2": "...", "task3": "..."}
[26 few-shot examples covering grief reactions, religious expressions,

political solidarity, admiration, and ambiguous praise]

The prompt was iteratively refined over 8 versions, improving Subtask 1 F1 from 0.616 (v1, 100-sample development subset) to 0.918 (v8, 100-sample development subset), and reaching 0.833 on a separate 200-sample evaluation subset.

Table 4

LLM judge (GPT-5.4-mini) Subtask 1 macro F1 across prompt iterations on a 100-sample development subset.

Version	Key Change	T1 F1
v1	Initial combined prompt	0.616
v2	Tightened Non-Supportive	0.789
v3	Added “admiration” rule	0.840
v5	Rebalanced examples	0.855
v7	Fixed admiration boundary	0.887
v8	Analytical text rule	0.918

The most impactful change was tightening the Non-Supportive definition (v1→v2, +17.3 F1 points): the initial prompt over-classified emotional reactions and generic praise as Supportive. Incorporating the annotation guideline that “admiration directed at a person” counts as Supportive (v2→v3) recovered false negatives without losing precision.

Hybrid: BERTweet + LLM verification. We additionally experiment with a hybrid approach where BERTweet predictions are provided to the LLM as “pre-labels” alongside the original text. The LLM is instructed to either confirm or correct the BERTweet prediction. We find that this approach suffers from anchoring bias: the LLM agrees with the pre-label 88–97% of the time depending on the subtask, limiting its ability to correct errors.

6. Results

We report three evaluation regimes. First, internal validation results are computed from stratified 80/10/10 splits of the official training data. Baseline runs use seed 42; transformer experiments use seeds 13, 42, and 77, with best-seed and mean/std results reported where available. Second, LLM prompt-development results are measured on 100- and 200-sample internal subsets due to API cost. Third, official CodaBench scores are leaderboard scores on the hidden test set and are not directly comparable to internal per-subtask F1.

6.1. Official CodaBench Results

Table 5 reports the official CodaBench scores from the SSD-2026 leaderboard. These scores are macro-averaged leaderboard scores for a complete language submission and should be distinguished from the internal per-subtask validation results in Tables 6 and 7.

Table 5

Official CodaBench scores for submitted systems.

Submission	Lang.	Strategy	Score
submission_english.zip	EN	Initial English pipeline	0.8231
submission_english_v2.zip	EN	English pipeline v2	0.8231
submission_english_bertweet.zip	EN	BERTweet-only pipeline	0.8231
submission_english_full.zip	EN	Full pipeline	0.8201
submission_en_hybrid_bertweet_v2.zip	EN	BERTweet+LLM hybrid	0.8016
submission_english_llm.zip	EN	LLM judge pipeline	0.7918
submission_spanish_full.zip	ES	Full pipeline	0.8785
submission_spanish.zip	ES	RoBERTuito-only pipeline	0.8544
submission_spanish_v3.zip	ES	Spanish v3 pipeline	0.8459
submission_es_hybrid_v3.zip	ES	Spanish hybrid pipeline	0.7935

Three English submissions tied at the best official English score of 0.8231. Our full English pipeline scored 0.8201, marginally below the tied best; adding XLM-R multilingual for Subtask 3 improved internal minority-class behavior but did not improve the aggregate official score, possibly because it over-predicted minority classes on the official test distribution. The Spanish full pipeline achieved the best Spanish score, 0.8785.

6.2. English Results

Table 6 presents the full English validation results with macro-averaged precision, recall, and F1 for the best seed of each model. For transformer models, we additionally report the mean F1 $\pm$ standard deviation across three random seeds.

Table 6

English validation results (best seed). Macro-averaged Precision (P), Recall (R), and F1. For transformers, mean F1 $\pm$ std over 3 seeds in the rightmost column. **Bold** = best F1 per subtask among fine-tuned models; underline = overall best including LLM.

Approach	Model	Subtask 1			Subtask 2			Subtask 3			Mean F1
		P	R	F1	P	R	F1	P	R	F1	
Trad. ML	LR + TF-IDF	.789	.823	.803	.756	.806	.775	.706	.742	.721	—
	SVM + TF-IDF	.761	.806	.778	.831	.568	.573	.723	.738	.729	—
Deep Learn.	TextCNN	.759	.826	.778	.857	.808	.829	.861	.869	.836	.776 $\pm$.002
	BiLSTM	.778	.837	.798	.813	.819	.816	.735	.764	.731	.788 $\pm$.007
Single-Task	BERTweet	.807	.860	.831	.877	.919	.896	.703	.764	.764	.826 $\pm$.005
	XLM-R (EN)	.781	.839	.801	.816	.866	.836	.827	.782	.789	.792 $\pm$.007
XLM-R (Multi)	XLM-R	.811	.856	.829	.840	.851	.846	.818	.921	.858	.847 $\pm$.009
Multi-Task	XLM-T (Multi)	.772	.837	.793	.900	.905	.903	.814	.901	.847	—
	XLM-T (EN)	.757	.821	.777	.876	.814	.841	.750	.743	.746	—
LLM Judge	GPT-5.4-mini	.822	.845	<u>.833</u>	.857	.857	.857	.946	.944	<u>.944</u>	—

Subtask 1. BERTweet achieves the highest fine-tuned F1 of 0.831 ($P = 0.807$, $R = 0.860$), benefiting from domain-specific pretraining on English tweets. Its high recall indicates strong sensitivity to supportive language, while precision is slightly lower due to borderline cases (e.g., emotional reactions misclassified as supportive). The LLM judge reaches a comparable 0.833 on an internal 200-sample evaluation subset ($P = 0.822$, $R = 0.845$), though this is not directly comparable due to the smaller evaluation set. XLM-R multilingual (0.815 mean) outperforms XLM-R English-only (0.792), suggesting that Spanish training data provides useful cross-lingual signal for support detection. Traditional ML baselines remain competitive (LR: 0.803), demonstrating that TF-IDF features capture substantial signal for binary support classification.

Subtask 2. The multi-task XLM-Twitter model achieves the highest Subtask 2 F1 of 0.903 ($P = 0.900$, $R = 0.905$), with well-balanced precision and recall, suggesting that joint training with Subtasks 1 and 3 provides useful inductive bias for target type classification. BERTweet’s best single seed reaches 0.896 ($P = 0.877$, $R = 0.919$) with notably higher recall, though with greater variance across seeds ($\text{std} = 0.032$).

Subtask 3. On the internal LLM evaluation subset, the LLM judge achieves the highest Subtask 3 F1 of 0.944 ($P = 0.946$, $R = 0.944$) by correctly classifying minority categories (Religion, Women) that fine-tuned models struggle with due to limited training data (15 and 19 examples respectively). Among fine-tuned models, XLM-R multilingual ($F1 = 0.858$, $P = 0.818$, $R = 0.921$) significantly outperforms BERTweet ($F1 = 0.764$), with its advantage concentrated in the underrepresented classes. The high recall of XLM-R multilingual (0.921) reflects the benefit of additional Spanish training examples for minority classes. BERTweet predicts Religion for only 1 out of 2,000 test samples, reflecting the extreme sparsity of this class in English-only training data.

6.3. Spanish Results

Table 7 presents the Spanish validation results.

Table 7

Spanish validation results (best seed). Macro-averaged Precision (P), Recall (R), and F1. For transformers, mean $F1 \pm \text{std}$ over 3 seeds in the rightmost column.

Approach	Model	Subtask 1			Subtask 2			Subtask 3			Mean F1
		P	R	F1	P	R	F1	P	R	F1	
Trad. ML	LR + TF-IDF	.823	.831	.827	.841	.824	.832	.831	.857	.822	—
	SVM + TF-IDF	.872	.759	.796	.807	.632	.655	.738	.671	.692	—
Single-Task	RoBERTuito	.823	.873	.865	.901	.953	.953	.974	.974	.974	.843 $\pm$.022
	XLM-R (ES)	.857	.904	.877	.770	.741	.753	.715	.795	.736	.836 $\pm$.031
Multi-Task	XLM-T (ES)	.810	.874	.832	.818	.828	.823	.892	.917	.901	—

RoBERTuito dominates the Spanish results on Subtasks 2 and 3, achieving F1 scores of 0.953 ($P = 0.901$, $R = 0.953$) and 0.974 ($P = R = 0.974$) respectively. The near-perfect precision and recall on Subtask 3 confirm the advantage of domain-specific pretraining: RoBERTuito was pretrained on 500M+ Spanish tweets, giving it strong prior knowledge of informal Spanish text patterns. For Subtask 1, RoBERTuito reaches 0.865 with higher recall (0.873) than precision (0.823), indicating it captures most supportive comments but occasionally over-predicts.

XLM-R shows high instability on Spanish, particularly for Subtask 3 ($\text{std} = 0.196$), with one seed collapsing to $F1 = 0.276$. This suggests that the smaller Spanish training set (2,600 samples) is insufficient to reliably fine-tune a general-purpose multilingual model for fine-grained community classification.

Notably, XLM-R ES achieves the highest single-seed Subtask 1 F1 (0.877, $P = 0.857$, $R = 0.904$), outperforming RoBERTuito’s best (0.865). For the final submission, we use XLM-R for Spanish Subtask 1 and RoBERTuito for Subtasks 2 and 3, selecting the most reliable model for each.

6.4. Cross-Lingual and Multilingual Analysis

Training XLM-R on combined English + Spanish data (“multilingual”) consistently improves English results compared to English-only training: Subtask 1 rises from 0.792 to 0.815, Subtask 2 from 0.821 to 0.832, and Subtask 3 from 0.722 to 0.847. The most dramatic improvement is on Subtask 3, where the additional Spanish training data for underrepresented classes (e.g., Religion: 15 EN + 16 ES) alleviates extreme data sparsity.

6.5. Hybrid Approach: BERTweet + LLM Verification

In the hybrid approach, BERTweet pre-labels are shown to the LLM with the instruction to confirm or correct them. On the English test set (2,000 samples), the LLM changed 145 Subtask 1 labels (7.25%), 43 Subtask 2 labels, and 32 Subtask 3 labels. The high agreement rate (93–97%) indicates strong anchoring bias: the LLM tends to defer to the suggested label rather than independently evaluate. This limits the hybrid approach’s ability to correct systematic errors in the pre-labels.

6.6. Error Analysis

Class imbalance effects. All fine-tuned models struggle with the Religion class (15 training examples). BERTweet achieves 0.0 F1 for Religion across most seeds, while XLM-R multilingual reaches 0.82 by leveraging Spanish Religion examples. The LLM judge, requiring no training data, correctly classifies Religion examples via its world knowledge.

Figure 8 shows the confusion matrices for the best BERTweet Task 1 and Task 2 English models. The main Task 1 errors are Supportive comments predicted as Non-Supportive, consistent with the difficult boundary between emotional commentary and directed care illustrated in Table 1. All confusion matrices are computed from the internal held-out test split of the corresponding best-seed artifact: BERTweet seed 13 for English Subtasks 1–2, BERTweet seed 77 for English Subtask 3, XLM-R multilingual seed 13 for English Subtask 3, XLM-R seed 42 for Spanish Subtask 1, and RoBERTuito seed 42 for Spanish Subtask 3.

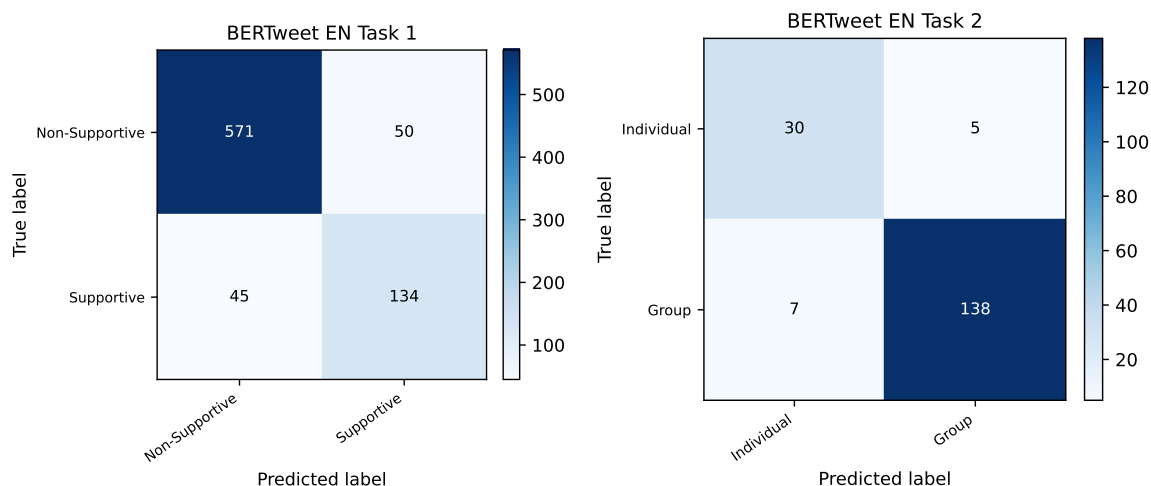


Figure 8: Confusion matrices for BERTweet on English Subtasks 1 and 2.

Figure 9 contrasts BERTweet and multilingual XLM-R for English Subtask 3. The multilingual model improves recall for underrepresented communities, including Religion and Women, which supports the internal validation result that multilingual training improves minority-class behavior.

Figure 10 shows the strongest Spanish components: XLM-R for Subtask 1 and RoBERTuito for Subtask 3. RoBERTuito’s cleaner Subtask 3 matrix reflects the benefit of Spanish social-media pretraining for fine-grained community labels.

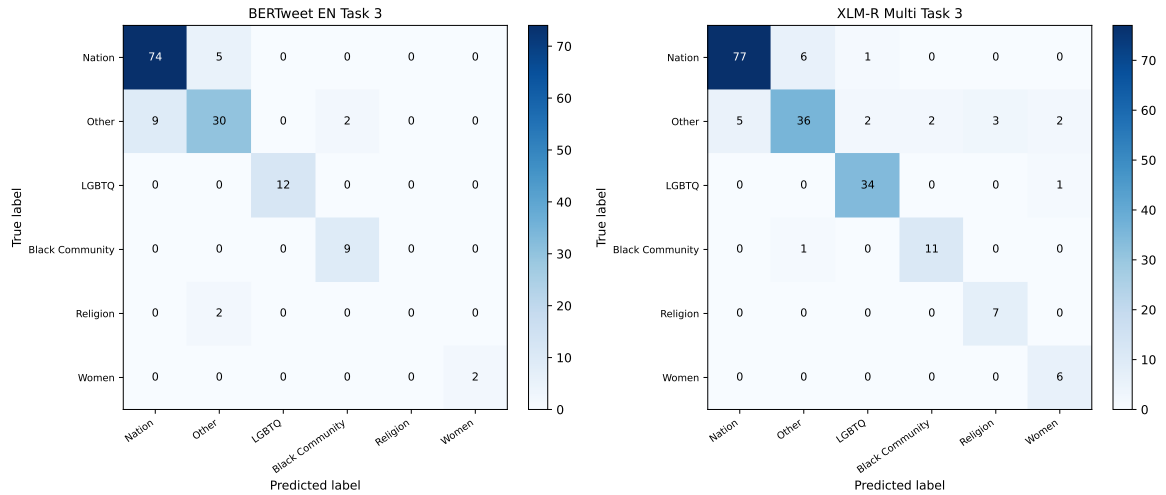


Figure 9: English Subtask 3 confusion matrices: BERTweet (left) versus multilingual XLM-R (right).

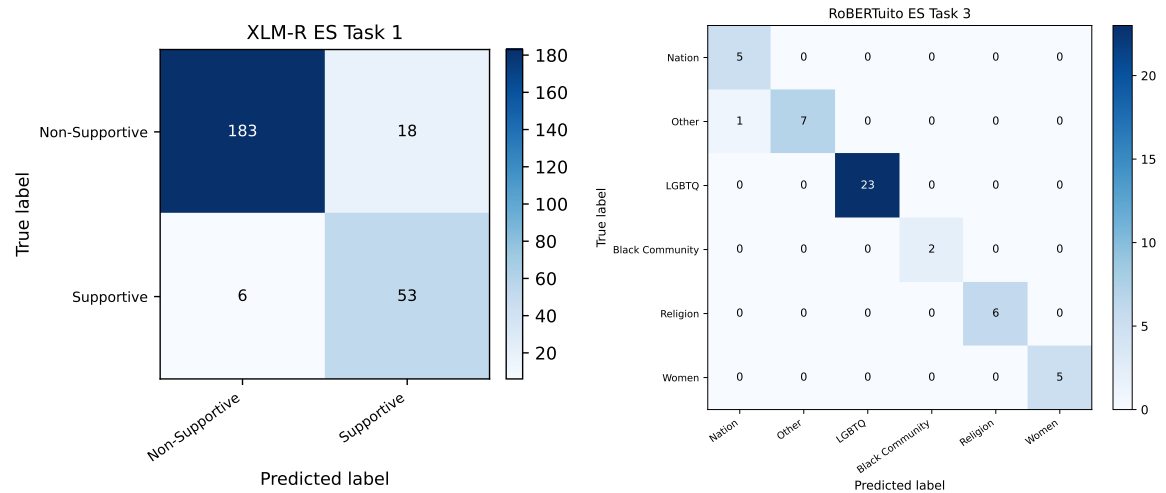


Figure 10: Confusion matrices for Spanish Subtask 1 with XLM-R (left) and Subtask 3 with RoBERTuito (right).

Boundary cases. The most challenging cases for Subtask 1 involve distinguishing directed support from emotional reactions. Comments like “tears come to my eyes watching what’s happening in Palestine” express emotion about a situation but do not direct support at anyone — a nuance that required explicit prompt engineering to capture.

Seed sensitivity. Transformer models show notable seed sensitivity, particularly on smaller subtask datasets. BERTweet Subtask 2 ranges from 0.819 to 0.896 across seeds; XLM-R ES Subtask 3 ranges from 0.276 to 0.736. This underscores the importance of multi-seed evaluation for reliable model comparison.

7. Future Work and Limitations

The main limitation of the LLM-as-judge experiments is evaluation scale. Internal prompt iterations were measured on 100–200-sample subsets because API cost made full repeated evaluation impractical, and the official LLM-only submission score (0.7918) is lower than the strongest internal subset results. The hybrid BERTweet+LLM strategy is also expensive while providing limited benefit because the LLM tends to anchor on the supplied pre-label.

The dataset itself imposes additional limits. Extreme class imbalance makes minority-class estimates

unstable: in English, Religion has only 15 training examples and Women has 19, so a small number of errors can substantially shift macro F1. Spanish contains 2,600 training comments, roughly one third of the English training size, which limits robust cross-lingual generalization despite the strong performance of RoBERTuito. Because Subtask 3 assigns support to sensitive community categories such as LGBTQ, Religion, Women, Black Community, and Nation, deployment requires care. Misclassification may produce different reliability across communities, especially for rare classes, and systems of this kind could be misused for surveillance of political, religious, or identity-based discourse. We therefore view the submitted systems as shared-task classifiers rather than deployment-ready moderation or monitoring tools.

Future work should explore curriculum learning that first stabilizes majority classes and then gradually emphasizes minority communities, augmentation for rare classes through paraphrasing or back-translation, and larger multilingual encoders such as multilingual DeBERTa-v3. For LLM-based systems, a more promising hybrid design would combine transformer confidence scores and LLM judgments without exposing the candidate label directly to the LLM. Finally, the emotion and sentiment signals identified here — solidarity phrases, faith-based language, and political support discourse — should be tested on other languages and platforms.

8. Conclusion

We presented a comprehensive comparison of five modeling paradigms for the SSD-2026 social support detection task. Our key findings are:

1. **Domain-adapted transformers excel:** BERTweet (English) and RoBERTuito (Spanish) consistently outperform general-purpose models, confirming the value of pretraining on in-domain social media text.
2. **Multilingual training helps:** XLM-R trained on combined English + Spanish data improves English Subtask 3 by 12.5 F1 points over English-only training, particularly for minority classes.
3. **LLM-as-judge is strong on minority classes:** GPT-5.4-mini achieves 0.944 on an internal Subtask 3 evaluation subset without any training data, correctly handling classes where fine-tuned models fail due to data sparsity.
4. **Prompt engineering matters:** Eight iterations of prompt refinement improved Subtask 1 F1 by 30 points, with the annotation guideline’s treatment of “admiration” being a critical insight.
5. **Hybrid approaches face anchoring bias:** Providing pre-labels to the LLM causes it to over-agree (93–97%), limiting correction gains.

On the official CodaBench leaderboard, our best submissions achieved 0.8231 for English and 0.8785 for Spanish. These results show that domain-adapted transformer pipelines remain highly competitive, while emotion-aware LLM prompting is useful for analysis and minority-class diagnosis.

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